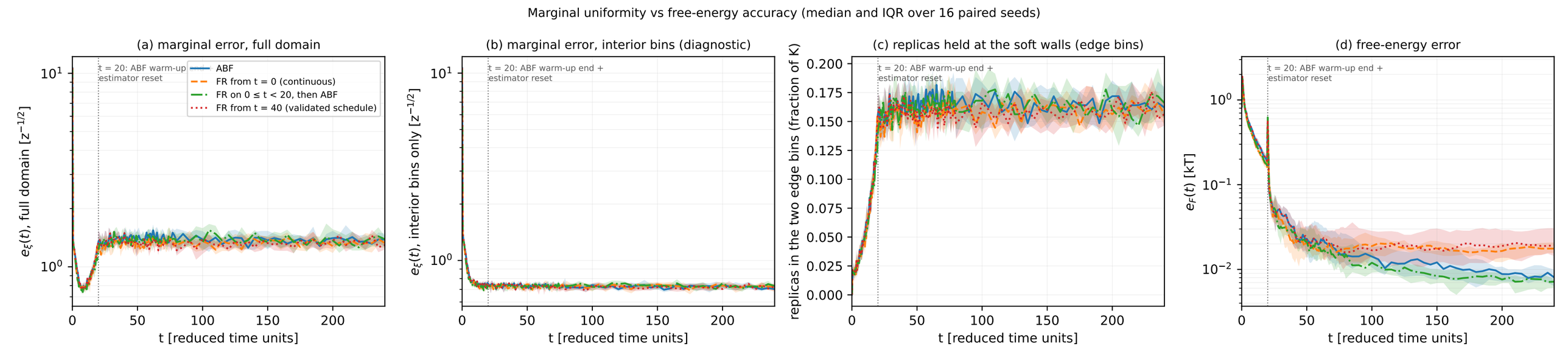

Marginal uniformity vs free-energy accuracy (median and IQR over 16 paired seeds)
(a) marginal error, full domain
(b) marginal error, interior bins (diagnostic)
(c) replicas held at the soft walls (edge bins)
(d) free-energy error
t = 20: ABF warm-up + estimator reset
ABF
FR from t = 0 (continuous)
FR on 0 ≤ t < 20, then ABF
FR from t = 40 (validated schedule)
$e_\xi(t)$, full domain [$z^{-1/2}$]
$e_\xi(t)$, interior bins only [$z^{-1/2}$]
replicas in the two edge bins (fraction of K)
$e_F(t)$ [kT]
t [reduced time units]